

**SIMATS ENGINEERING
ASSESSMENT TOOL 4**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 20%

QUESTIONS: 5

**Time Duration: 1 Hour
Total Marks:30**

Annexure A

Resolution Algorithm Implementation

Code of Conduct:

I _____ HARIHARAN,R(192524003) _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:HARIHARAN.R

Annexure A

Resolution Algorithm Implementation

Question 1 – Rain and Wet Ground

Knowledge Base

1. If it rains, then the ground becomes wet.
2. It is raining.

Goal

Prove that **the ground is wet** using the Resolution Algorithm.

Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF.

Negate the goal.

1. Apply the Resolution Algorithm step by step.
2. Show the Empty Clause ($\square$) if the goal is proved.

Question 2 – Student Assignment Submission

Knowledge Base

1. If a student submits the assignment, then the student receives internal marks
2. Rahul submitted the assignment.

Goal

Prove that **Rahul receives internal marks** using the Resolution Algorithm.

Tasks

1. Write the propositional logic expressions.
2. Convert the premises into CNF.

Negate the goal.

1. Apply the Resolution Algorithm.
2. Conclude whether the goal is proved.

Question 3 – Library Membership

Knowledge Base

1. If a person is a library member, then the person can borrow books.
2. Priya is a library member.

Goal

1. Prove that **Priya can borrow books** using the Resolution Algorithm.

Tasks

1. Represent the knowledge base using propositional logic.
2. Convert each statement into CNF.

Negate the goal.

1. Perform resolution until no new clauses can be generated.
2. State the final conclusion.

Question 4 – Placement Eligibility

Knowledge Base

1. If a student clears the aptitude test, then the student is eligible for placement.
2. Arun cleared the aptitude test.

Goal

1. Prove that **Arun is eligible for placement** using the Resolution Algorithm.

Tasks

1. Convert the statements into propositional logic.
2. Write the CNF clauses.
3. Negate the goal.
4. Apply the Resolution Algorithm step by step.
5. Draw the resolution tree and conclude.

Question 5 – Access Control System

Knowledge Base

1. If a user enters the correct password, then the user is authenticated.
2. If a user is authenticated, then the user is granted access.
3. The user entered the correct password.

Goal

Prove that **the user is granted access** using the Resolution Algorithm.

Tasks

1. Represent the knowledge base in propositional logic.
2. Convert all implications into CNF.
3. Negate the goal.
4. Apply the Resolution Algorithm to derive the Empty Clause ($\square$).
5. Explain each resolution step and write the final conclusion.

ANSWER::

RESOLUTION ALGORITHM – PROPOSITIONAL LOGIC

QUESTION 1 – RAIN AND WET GROUND

Knowledge Base

1. If it rains, then the ground becomes wet.
2. It is raining.

Goal

Prove that **the ground is wet** using the Resolution Algorithm.

1. Represent the Statements in Propositional Logic

Let:

- **R** = It is raining.
- **W** = The ground is wet.

The statements are:

1. If it rains, then the ground becomes wet.

$R \rightarrow W$

2. It is raining.

R

2. Convert the Statements into CNF

We know:

$R \rightarrow W$

An implication can be converted as:

$\neg R \vee W$

Therefore, the CNF clauses are:

- **Clause 1: $\neg R \vee W$**
- **Clause 2: R**

3. Negate the Goal

Goal:

W

Negated goal:

$\neg W$

Add the negated goal to the knowledge base.

Therefore, the clauses are:

1. **$\neg R \vee W$**
2. **R**
3. **$\neg W$**

4. Apply Resolution Algorithm

Resolve Clause 1 and Clause 2:

- Clause 1: **$\neg R \vee W$**
- Clause 2: **R**

The complementary literals are **$\neg R$** and **R**.

Resolving them gives:

W

So we obtain:

Clause 4: W

Now resolve Clause 4 with the negated goal:

- Clause 4: **W**
- Clause 3: **$\neg W$**

The complementary literals are **W** and **$\neg W$** .

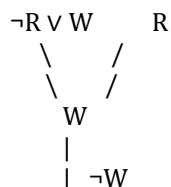
Therefore:

$W \vee \neg W \rightarrow \square$

Hence:

$\square$ = Empty Clause

5. Resolution Tree





6. Final Conclusion

Since the **Empty Clause** ($\square$) is derived, the negation of the goal leads to a contradiction. Therefore, the goal is proved.

Hence, the ground is wet.

QUESTION 2 – STUDENT ASSIGNMENT SUBMISSION

Knowledge Base

1. If a student submits the assignment, then the student receives internal marks.
2. Rahul submitted the assignment.

Goal

Prove that **Rahul receives internal marks** using the Resolution Algorithm.

1. Propositional Logic Expressions

Let:

- **S** = Rahul submitted the assignment.
- **M** = Rahul receives internal marks.

The statements are:

1. If Rahul submits the assignment, then Rahul receives internal marks.

$S \rightarrow M$

2. Rahul submitted the assignment.

S

2. Convert the Premises into CNF

Convert:

$S \rightarrow M$

Using:

$P \rightarrow Q \equiv \neg P \vee Q$

Therefore:

$\neg S \vee M$

The CNF clauses are:

- **Clause 1: $\neg S \vee M$**
- **Clause 2: S**

3. Negate the Goal

Goal:

M

Negated goal:

$\neg M$

Add it to the clauses.

Therefore:

1. **$\neg S \vee M$**
2. **S**
3. **$\neg M$**

4. Apply Resolution Algorithm

Resolve Clause 1 and Clause 2:

- Clause 1: **$\neg S \vee M$**
- Clause 2: **S**

The complementary literals are **$\neg S$** and **S**.

Therefore:

M

We obtain:

Clause 4: M

Now resolve Clause 4 with Clause 3:

- Clause 4: **M**
- Clause 3: **$\neg M$**

The complementary literals are **M** and **$\neg M$** .

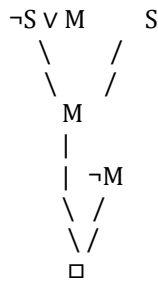
Therefore:

$M \vee \neg M \rightarrow \square$

Thus:

$\square$ = Empty Clause

5. Resolution Tree



6. Final Conclusion

The Empty Clause $\square$ is derived.

Therefore, the negation of the goal is contradictory.

Hence, Rahul receives internal marks.

QUESTION 3 – LIBRARY MEMBERSHIP

Knowledge Base

1. If a person is a library member, then the person can borrow books.
2. Priya is a library member.

Goal

Prove that **Priya can borrow books** using the Resolution Algorithm.

1. Represent the Knowledge Base Using Propositional Logic

Let:

- **L** = Priya is a library member.
- **B** = Priya can borrow books.

The statements are:

1. If Priya is a library member, then Priya can borrow books.

$L \rightarrow B$

2. Priya is a library member.

L

2. Convert Each Statement into CNF

Convert:

$L \rightarrow B$

Using:

$P \rightarrow Q \equiv \neg P \vee Q$

Therefore:

$\neg L \vee B$

The CNF clauses are:

- **Clause 1: $\neg L \vee B$**
- **Clause 2: L**

3. Negate the Goal

Goal:

B

Negated goal:

$\neg B$

Therefore, the complete set of clauses is:

1. **$\neg L \vee B$**
2. **L**
3. **$\neg B$**

4. Perform Resolution

Resolve Clause 1 and Clause 2:

- Clause 1: **$\neg L \vee B$**
- Clause 2: **L**

The complementary literals are **$\neg L$** and **L** .

Therefore:

B

New clause:

Clause 4: B

Now resolve Clause 4 and Clause 3:

- Clause 4: **B**
- Clause 3: **$\neg B$**

The complementary literals are **B** and **$\neg B$** .

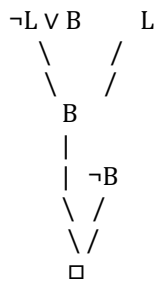
Therefore:

$B \vee \neg B \rightarrow \square$

Hence:

$\square$ = **Empty Clause**

5. Resolution Tree



6. Final Conclusion

The Empty Clause $\square$ is obtained.

Therefore, the negated goal $\neg B$ is false.

Hence, Priya can borrow books.

QUESTION 4 – PLACEMENT ELIGIBILITY

Knowledge Base

1. If a student clears the aptitude test, then the student is eligible for placement.
2. Arun cleared the aptitude test.

Goal

Prove that **Arun is eligible for placement** using the Resolution Algorithm.

1. Convert the Statements into Propositional Logic

Let:

- **A** = Arun cleared the aptitude test.
- **E** = Arun is eligible for placement.

The statements are:

1. If Arun clears the aptitude test, then Arun is eligible for placement.

$A \rightarrow E$

2. Arun cleared the aptitude test.

A

2. Write the CNF Clauses

Convert:

$A \rightarrow E$

Using:

$P \rightarrow Q \equiv \neg P \vee Q$

Therefore:

$\neg A \vee E$

The CNF clauses are:

- **Clause 1:** $\neg A \vee E$
- **Clause 2:** **A**

3. Negate the Goal

Goal:

E

Negated goal:

$\neg E$

Complete set of clauses:

1. $\neg A \vee E$
2. **A**
3. $\neg E$

4. Apply the Resolution Algorithm

Resolve Clause 1 and Clause 2:

- Clause 1: $\neg A \vee E$
- Clause 2: **A**

The complementary literals are $\neg A$ and **A**.

Therefore:

E

New clause:

Clause 4: E

Now resolve Clause 4 and Clause 3:

- Clause 4: **E**
- Clause 3: **¬E**

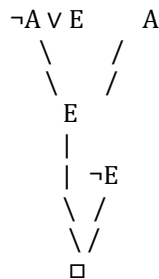
The complementary literals are **E** and **¬E**.

Therefore:

$$E \vee \neg E \rightarrow \square$$

Hence:

$\square$ = **Empty Clause**

5. Resolution Tree**6. Final Conclusion**

The Empty Clause $\square$ is derived.

Therefore, the negation of the goal produces a contradiction.

Hence, Arun is eligible for placement.

QUESTION 5 – ACCESS CONTROL SYSTEM**Knowledge Base**

1. If a user enters the correct password, then the user is authenticated.
2. If a user is authenticated, then the user is granted access.
3. The user entered the correct password.

Goal

Prove that **the user is granted access** using the Resolution Algorithm.

1. Represent the Knowledge Base in Propositional Logic

Let:

- **P** = The user entered the correct password.
- **A** = The user is authenticated.
- **G** = The user is granted access.

The statements are:

1. If the user enters the correct password, then the user is authenticated.
P → A
2. If the user is authenticated, then the user is granted access.
A → G
3. The user entered the correct password.
P

P

2. Convert All Implications into CNF

First implication:

$$P \rightarrow A$$

Using:

$$P \rightarrow Q \equiv \neg P \vee Q$$

Therefore:

$$\neg P \vee A$$

Second implication:

$$A \rightarrow G$$

Therefore:

$$\neg A \vee G$$

The CNF clauses are:

- **Clause 1: $\neg P \vee A$**
- **Clause 2: $\neg A \vee G$**
- **Clause 3: **P****

3. Negate the Goal

Goal:

G

Negated goal:

$\neg G$

Add the negated goal to the knowledge base.

Complete set of clauses:

1. $\neg P \vee A$
2. $\neg A \vee G$
3. P
4. $\neg G$

4. Apply the Resolution Algorithm

Resolution Step 1

Resolve Clause 1 and Clause 3:

- Clause 1: $\neg P \vee A$
- Clause 3: P

The complementary literals are $\neg P$ and P .

Therefore:

A

New clause:

Clause 5: A

Resolution Step 2

Resolve Clause 2 and Clause 5:

- Clause 2: $\neg A \vee G$
- Clause 5: A

The complementary literals are $\neg A$ and A .

Therefore:

G

New clause:

Clause 6: G

Resolution Step 3

Resolve Clause 6 and the negated goal Clause 4:

- Clause 6: G
- Clause 4: $\neg G$

The complementary literals are G and $\neg G$.

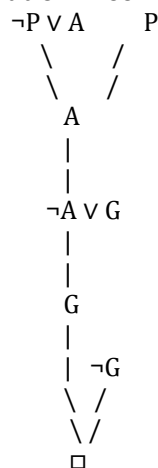
Therefore:

$G \vee \neg G \rightarrow \square$

Hence:

$\square$ = **Empty Clause**

5. Resolution Tree



6. Detailed Resolution Representation

Clause 1: $\neg P \vee A$

Clause 3: P

Clause 5: A

Clause 2: $\neg A \vee G$

Clause 5: A

Clause 6: G

Clause 6: G

Clause 4: $\neg G$

Clause 7: $\square$

7. Final Conclusion

The Empty Clause $\square$ is derived after applying the Resolution Algorithm.

This means that the negation of the goal $\neg G$ contradicts the knowledge base.

Therefore, the goal is proved.

Hence, the user is granted access.

OVERALL CONCLUSION

The Resolution Algorithm proves a goal by adding the **negation of the goal** to the knowledge base and repeatedly resolving clauses.

If the process produces the **Empty Clause ($\square$)**, a contradiction is obtained. Therefore, the original goal is logically entailed by the knowledge base.

The five goals are successfully proved:

1. **The ground is wet.**
2. **Rahul receives internal marks.**
3. **Priya can borrow books.**
4. **Arun is eligible for placement.**
5. **The user is granted access.**

In all five cases, the Resolution Algorithm derives the **Empty Clause ($\square$)**, proving that the respective goals are true.

RUBRICS FOR CALCULATION

Numerical Problems					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation